Harbani Kaur

Sophomore, Department of Computer Engineering

GitHub: harbani007 | **LinkedIn:** harbanikaur1 | **Email:** harbani.k1@gmail.com | **Phone:** +91-9870357566

Profile

Second-year B.E. Computer Engineering student focused on web development (HTML/CSS/JavaScript, React.js, Node.js) and building practical projects. Currently learning Machine Learning and strengthening problem-solving through data structures and algorithms.

Education

Thapar Institute of Engineering & Technology, Patiala — *2024 – 2028*

Bachelor of Engineering in Computer Engineering — *Patiala, India*

- **CGPA:** 9.15
- **Relevant Coursework:** Artificial Intelligence, Probability and Statistics, Numerical Linear Algebra, Discrete Mathematics, Data Structures, Operating Systems, Object-Oriented Programming

Amity International School, Pushp Vihar, New Delhi — *Schooling*

- **Class 10:** 98.4%
- **Class 12:** 91.2%

Projects

Attendance Management System (Face Recognition) | *Python, OpenCV, dlib, SQLite*

- Built an attendance system using a face recognition pipeline: face detection → alignment → 128-D face embeddings → identity matching using cosine/Euclidean distance with a tuned threshold.
- Implemented student enrollment (capture multiple samples per user), real-time webcam recognition, and attendance logging with timestamps; prevented duplicate marking within the same session/day.
- Stored only embeddings + IDs in a database (SQLite) and added basic image-quality checks (blur/low light) to reduce incorrect matches.

Email Spam Detection (NLP Classifier) | *Python, scikit-learn, NLP*

- Built an end-to-end spam classifier using text preprocessing + TF-IDF features and trained baseline models (Logistic Regression / Naive Bayes); packaged the workflow using a scikit-learn pipeline.
- Evaluated model performance using precision, recall, and F1-score; tuned n-grams and hyperparameters to reduce false positives (important for real inboxes).
- Added a simple UI/API to classify new email text and return prediction + confidence score.

Thapar Campus Class Navigator | *HTML, CSS, JavaScript*

- Built a campus class locator that maps timetable entries and lab codes to building/floor/room details, helping students find new labs and classrooms each semester.
- Displayed a campus map with interactive pins; selecting a result auto-focuses the map and opens a details panel with location information.
- Integrated a **Get Directions** feature that opens Google Maps navigation using the selected building's coordinates.

Technical Skills

- **Languages:** C/C++, JavaScript, HTML/CSS, Python
- **ML Libaries:** NumPy, Pandas, scikit-learn, OpenCV
- **Developer Tools:** Git/GitHub, VS Code, Linux, Kaggle
- **Mathematics for ML:** Linear Algebra (SVD, PCA), Probability & Statistics, Calculus

Certifications

- Machine Learning Specialization (DeepLearning.AI) — Coursera (Supervised, Advanced, Unsupervised/Recs/RL)
- Mastering Data Structures & Algorithms using C and C++ — Abdul Bari
- The Web Developer Bootcamp 2025 — Colt Steele
- NPTEL: Mathematical Foundations for ML (IISc Bangalore) — In Progress